San Francisco Bus Transit Access and Service Quality by ZIP Code

Project Documentation

github.com/mikaelskjonhaug/superplussenior_midterm

1 Research Question and Updates

Our original research question was: Among ZIP codes in San Francisco during a five-year period following COVID, is lower access to bus transit, measured by bus stops per square mile, associated with lower bus service quality, measured by average bus arrivals per stop and percentage of arrivals that are on time? Additionally, how does this relationship vary with ZIP-code socioeconomic status, measured using factors such as median household income and poverty rate? We wanted to determine whether there are measurable disparities in both transit access and service quality across San Francisco ZIP codes, and whether these disparities are associated with the socioeconomic characteristics of the areas being served.

Since developing our initial question, we reevaluated the definition of bus stop accessibility. Our plans post-midterm are to introduce walkability in terms of whether a sidewalk exists on one or both sides of the street surrounding a bus stop, and to investigate other potential proxies for accessibility, such as whether people walk or need to use a car to reach a stop, because distance can be difficult to calculate. To find proxies for the scarce ACS median-income data, which are aggregated over five years, we have also begun exploring ACS household income brackets through Dewey Data.

2 Methodology

2.1 Data sources and spatial joins

The ACS ZIP-level estimates came from the U.S. Census Bureau. We downloaded the Census ACS block-group income-bracket data through Dewey. ZIP boundaries came from DataSF, and block-group boundaries came from Census TIGER/Line files. The IRS income measures came from IRS Statistics of Income ZIP-code tables.

We assigned each ACS block-group centroid to a San Francisco ZIP polygon (666 of 681 matched) and joined SFMTA GTFS stops to ZIPs. For ZIP-area measures, we projected polygons to EPSG:7131 (NAD83(2011) / San Francisco CS13) and converted square metres to square miles. The street network came from OpenStreetMap through OSMnx.

The observed bus-arrival counts came from historical 511 regional transit data. Expected counts came from SFMTA GTFS schedule data assembled for the same months. The arrival analysis then joined the prepared tables by ZIP code and month.

2.2 Data cleaning

The historical 511 data include regional transit observations from multiple operators and types of service. We kept only SFMTA bus and trolleybus routes. We removed canceled, skipped, and missing observations, duplicate records for the same trip, date, and stop, and records with invalid dates. We also removed records outside the month being processed, observations that could not be assigned to a San Francisco ZIP code, and records without an observed arrival time. Arrival times were reported either as clock times or as seconds after midnight, so we converted both formats into seconds. We adjusted trips that crossed midnight so they would not appear to be several hours early or late. We considered an arrival on time when it occurred between 1 minute early and 5 minutes late. After cleaning, 416,666,791 observations remained. Of these, 340,834 did not have a scheduled arrival time. We kept these observations in the total arrival counts, but did not use them to calculate delays or on-time performance. This left 416,325,957 observations that could be compared with a scheduled time. Because each large 511 archive was summarized and deleted after its monthly file was created, we did not retain separate counts for every individual removal step.

We also cleaned the geographic, schedule, and income data before joining them. ZIP codes were stored as five-digit text values so the tables could be matched correctly. Stop coordinates were converted into map points and joined to San Francisco ZIP boundaries. ACS block groups were assigned to the ZIP code containing their center; 666 of 681 block groups matched, while 15 did not match a San Francisco ZIP. Household counts were then summed by ZIP code and year. We did not replace the missing ACS median household income for ZIP 94104 because the IRS data measure tax returns rather than households. This left 55 ZIP-month rows without a median income value. For the final dataset, we created every combination of 27 ZIP codes and 55 months before joining the observed arrivals, scheduled service, transit access, and demographic data. This produced 1,485 ZIP-month rows with no duplicate or missing ZIP-month combinations. We kept months with limited 511 coverage but marked them with data-quality flags. February 2023 failed the service-day check, affecting 27 rows, and November 2024 through February 2025 failed the citywide arrival-volume check, affecting 108 rows. In total, 1,350 rows passed both checks and 135 rows were flagged but retained. We felt that imputation would be

overly complex or not possible, so we note this as a limitation.

2.3 Income and socioeconomic status

The dataset builder combines ACS ZIP-level data, ACS block-group income brackets, and IRS Statistics of Income data for the single analysis notebook. ZIP codes were stored as five-digit text values so the datasets could be joined. We assigned ACS block groups to ZIP codes using their geographic centroids, then summed household counts in each income bracket by ZIP and year. The map uses the 2024 ACS estimates.

For each ZIP, we calculated the share of households earning under $25,000 and the share earning over $100,000. We ranked ZIPs relative to the other San Francisco ZIPs on both measures. The low-income rank was reversed, and the two ranks were averaged to make a composite socioeconomic score. We divided the scores into thirds to label ZIPs low, medium, or high. These labels are relative to San Francisco rather than fixed income cutoffs.

ACS medians and IRS measures were kept as additional features. IRS data cover tax years 2020–2022 and are reserved for later modeling. Prepared ACS income-bracket shares are used in the midterm map and income results. The map overlays ZIP boundaries, streets, and GTFS stops.

We retained ACS median household income as a dollar reference in the map legend where it is available. ZIP 94104 has no ACS median estimate, but it does have the income-bracket shares needed for its map score. We did not fill the missing median with IRS data: IRS data describe tax returns rather than households and cover earlier tax years, 2020–2022. The map colors are based only on the ACS bracket-share score.

2.4 Arrival divergence

For arrival divergence, the combined notebook compares observed 511 arrival counts with expected SFMTA scheduled-arrival counts for buses by ZIP and month. The shared data cover 27 ZIP codes and 55 months, from March 2022 through September 2026.

For each month, we divided a ZIP's arrivals by the total arrivals across the 27 ZIPs. This gives each ZIP an observed share and a scheduled share, even though the two datasets have different total counts.

We calculated KL divergence separately for each ZIP. In each month, we compared the observed and scheduled chances that an arrival was in that ZIP rather than elsewhere in the study area. We averaged these monthly KL values to get one score per ZIP. A larger score means its observed shares differ more from its scheduled shares.

We also ran a separate two-sample KS test for each ZIP, comparing its 55 observed monthly shares with its 55 scheduled monthly shares. The KS statistic is the largest gap between their empirical cumulative distribution functions (CDFs). We report each test's p-value and flag ZIPs with $p < 0.05$. The plots show the CDFs and their largest gaps.

2.5 Bus access and service quality

The access and service analysis uses the cleaned ZIP-month dataset, keeping only the 1,350 ZIP-months that passed the data-quality check. We measured access as bus stops per 1,000 residents, using population from the 2024 ACS estimates. This shows how many stops are available relative to the number of people they serve.

For each ZIP, we calculated two measures of service quality. Frequency is the number of bus arrivals per stop per day. On-time performance is the percentage of arrivals between 1 minute early and 5 minutes late. We summed the counts across months before dividing, so months with more data count more. ZIP 94104 (Financial District) was excluded because it has only about 500 residents and no ACS median income estimate, leaving 26 ZIPs. Our charts use ACS median household income; the bracket shares ranked ZIP codes almost the same way (Spearman $\rho = -0.93$ between median income and the share of households under $50,000). We used Spearman correlation because the data had outliers.

3 Results

Using 2024 ACS household income brackets, we group San Francisco ZIP codes into three relative income groups based on their shares of households earning below $25,000 and above $100,000. Physical transit stops are overlaid to show their locations; this map does not assess service quality. The legend reports the range of ZIP-level median household incomes within each group.

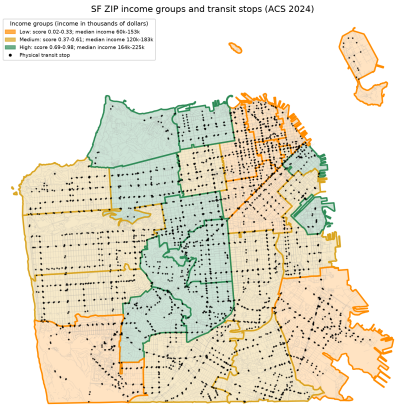


Figure 1: San Francisco ZIP codes are colored by relative socioeconomic group based on ACS income-bracket shares; black dots show physical transit stops.

Each box shows how the percentage of households in an income bracket varies across San Francisco ZIP codes. The box contains the middle 50% of ZIP percentages, and its center line marks the median. Whiskers extend to values within 1.5 times the interquartile range. Labeled orange dots identify ZIPs with unusually high or low household shares in that bracket.

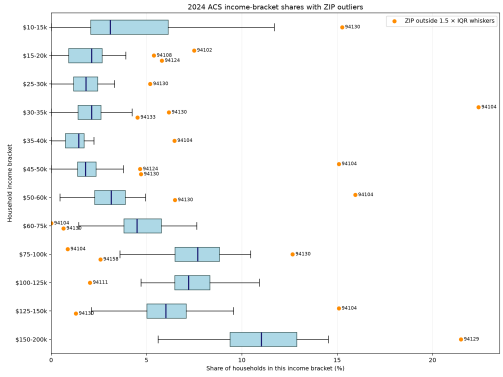


Figure 2: Box plots show the distribution of ZIP-level household shares across income brackets; labeled orange points identify outlier ZIP codes.

Eight ZIPs have KS p-values below 0.05: 94102, 94104, 94105, 94118, 94122, 94132, 94134, and 94158. ZIP 94158 is shown because it has the smallest KS p-value.

These results describe differences in monthly arrival shares, not whether buses arrived early or late. The p-values are exploratory: we ran 27 tests, and observations from different months may be related. ZIPs with low p-values show evidence that the observed and expected arrival-share distributions differ. The per-ZIP average observed-minus-scheduled share is shown in Figure 4.

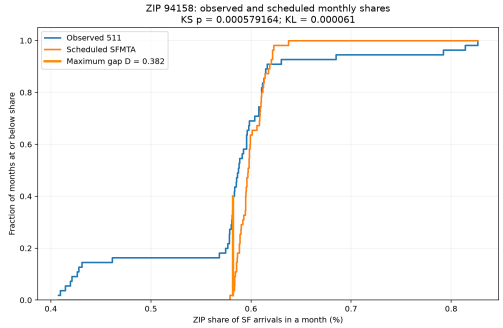


Figure 3: Observed and scheduled monthly arrival-share CDFs for ZIP 94158 show the largest KS gap for this ZIP; the plot also reports its KL divergence.

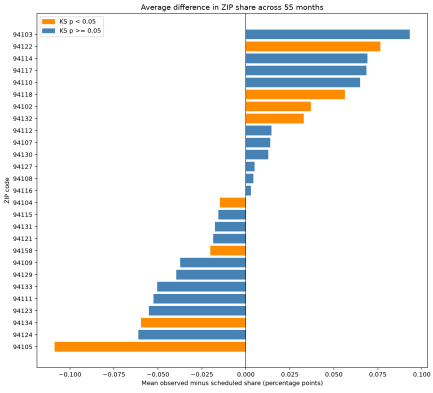


Figure 4: Average observed-minus-scheduled arrival share varies by ZIP across 55 months; orange bars mark ZIPs with exploratory KS p-values below 0.05.

ZIPs with fewer stops per resident had more arrivals per stop (Spearman $\rho = -0.68$). On-time performance showed little association with median income ($\rho = 0.07$); most ZIPs had 49%–60% of arrivals on time.

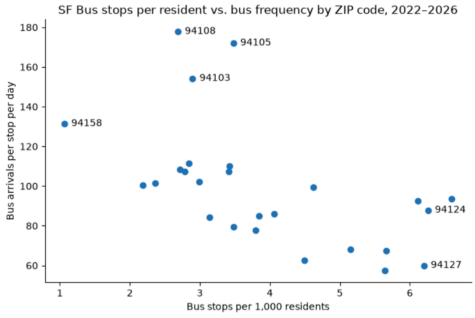


Figure 5: Fewer stops per resident correspond to more arrivals per stop (Spearman $\rho = -0.68$, $n = 26$).

Shared downtown stops may explain the negative stop-density association. On-time shares were similar across income groups.

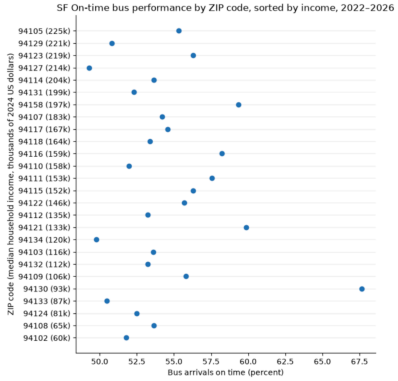


Figure 6: On-time performance shows little association with median household income (Spearman $\rho = 0.07$, $n = 26$).

Feedback requested. We seek advice on (1) stops per 1,000 residents versus walkable access and (2) OLS checks for 26 ZIPs, flagged months, and spatial dependence.

Next steps (targets). Tila Tran—validate ZIP/stop joins and flags (Nov. 10); Sharlena Nguyen—finish walkability proxy and ZIP features (Nov. 13); Christopher Mach—fit OLS and check confidence intervals and residuals (Nov. 18); Mikael Skjonhaug—finalize maps, charts, and outlier checks (Dec. 3).

Risks and adaptations. Missing ACS median income for 94104 and 135 flagged ZIP-months may bias estimates. We will use bracket shares for 94104, analyze passing months first, and check sensitivity to the flags. Monthly summaries keep the 511 archives manageable. If sidewalk coverage or OLS checks fail, we will report stops per resident and rank correlations with their limits stated.